

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458121.751 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.067 +/- 0.05
Transit depth (Rp/Rs)^2	0.44 +/- 0.66 [%]
Orbital Inclination (inc)	82.09 +/- 2.66
Airmass coefficient 1 (a1)	1.17 +/- 0.02
Airmass coefficient 2 (a2)	-0.0958 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.7 %
Best Comparison Star	None
Optimal Aperture	4.76
Optimal Annulus	9.92
Transit Duration (day)	0.03 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.067 ± 0.05 vs 0.1404 (deviation 0.93σ)
Duration fit vs published	0.03 d vs 0.0754 d (deviation 0.92σ)
Mid-time O–C	16.6 min
Depth SNR	0.8
Residual scatter	1.7 %

Light curve

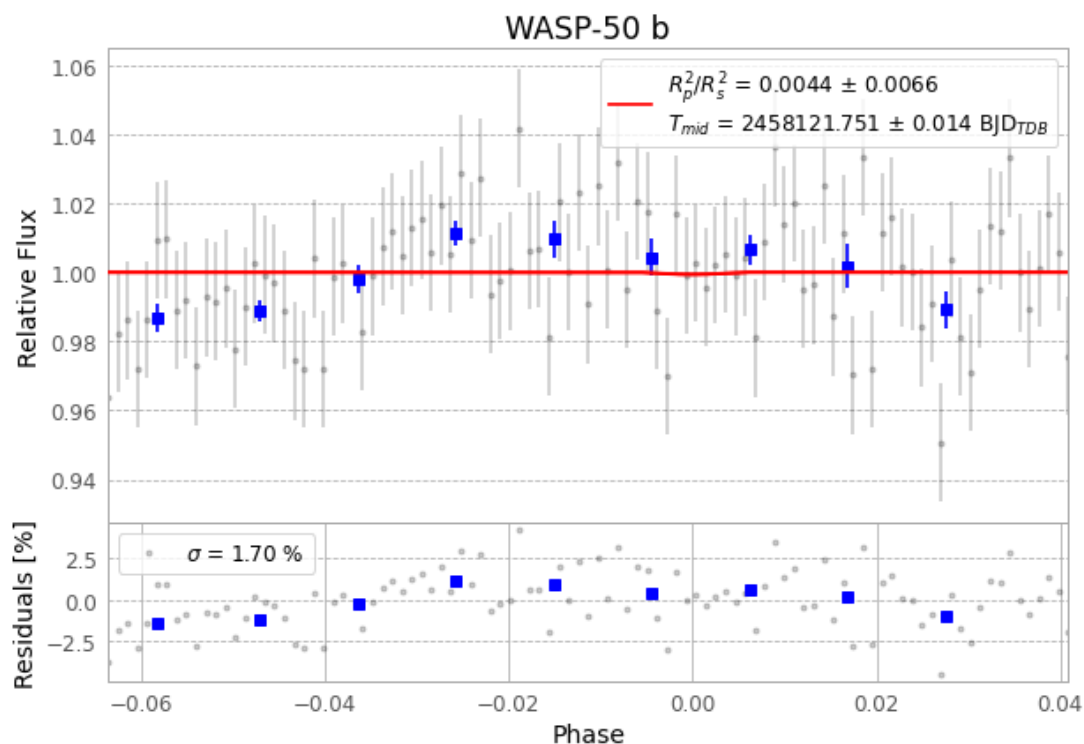


Figure 1 — FinalLightCurve_WASP-50 b_2018-01-02.png (SHA-256 1fde32a8e4b9c7f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3a2a6d8d129e25f4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

104 FITS frames (68 MB), WASP-50180103025712.FITS → WASP-50180103080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-180103.log (2cd2ea37d366...), FinalLightCurve_WASP-50 b_2018-01-02.png (1fde32a8e4b9...), FinalLightCurve_WASP-50 b_2018-01-02.csv (d5fab4a08c46...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 22:56Z.